3D print job: Pi Touch Display 2 satellite stand

4 parts, 1 of each Customer: _____ Needed by: _____

Colours. Frame: Spectrum PET-G MATT **Olive Green** (RAL 6003), or the closest matte olive/forest green the customer approves (prints at 230–255 °C nozzle / 60–80 °C bed; the 240 / 80 °C in the project files fits). **Stand, speaker box, cover:** _____

Print every file exactly as supplied. Don't rotate or scale. Each STL is already in its print orientation, in millimetres, at 100%. The parts snap and screw together, so sizes matter.

Parts

File	Qty	Size (mm)	Sits on the bed	Supports	Brim	Est. PETG / time*
stand.stl	1	190 × 123 × 82	Open front face down	**None**	6 mm	159 g / 14 h 50 m
frame.stl	1	194 × 125 × 16	Front lip down (visible face)	**None**	No	25 g / 2 h 07 m
speaker-box.stl	1	80 × 80 × 60	Open rim down	**None**	No	60 g / 5 h 35 m
pi-ports-cover.stl	1	64 × 42 × 3.6	Flat plate down (visible face)	**None**	No	3 g / 20 m
					Total	**≈ 246 g / 23 h**

* From PrusaSlicer 2.7 with the settings below, at conservative speeds. A fast printer will be quicker. **Bed needed: at least 200 × 130 mm** for the stand and frame. The speaker box and cover fit on one plate together.

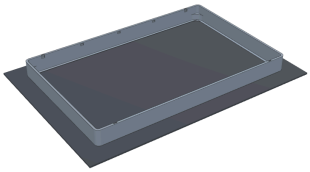
Settings (all parts)

Material	**PETG** recommended: a Raspberry Pi runs warm inside the stand. PLA+ is acceptable in a cool room. ASA is fine too.
Nozzle / layer height	0.4 mm / 0.20 mm (0.16 mm is fine for a nicer finish)
Walls	**4 perimeters** (the walls are 1.8 mm thick, so they print solid)
Top / bottom	5 top layers / 4 bottom layers
Infill	25% gyroid (any pattern is fine)
Supports	**OFF for all parts.** Every part was slice-checked and prints without supports. Supports inside the stand or speaker box could not be removed.
Texture (frame only)	**Fuzzy skin on the frame, outside walls only**: thickness 0.25 mm, point distance 0.7 mm (already set in plate2-frame.3mf). This textures the outside band only; the inside faces (glass pocket, snap grooves) stay smooth. **No fuzzy skin on the stand, speaker box or cover.**
Slicer options	Leave "detect thin walls" **off**. Apply elephant's-foot compensation of 0.1–0.2 mm: the stand's front rim and the frame's front lip are snap-fit faces.
Ready-made project files	**plate1-stand.3mf**, **plate2-frame.3mf** and **plate3-speaker-box-and-cover.3mf** open in PrusaSlicer with all of the above already set (PETG at 240 / 80 °C, 250 × 210 mm bed). Other slicers import the geometry; apply the settings in this table. **petg-profile.ini** holds the same settings.

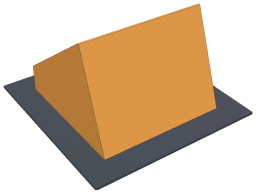
How each part sits on the bed



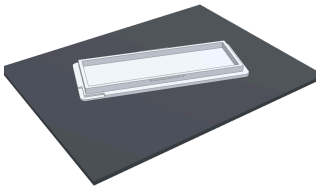
stand: open front face down (as the original design prints). Holes and engraving face up.



frame: front lip down. That face shows, so the plate texture is the customer's choice. Outside band: fuzzy skin.

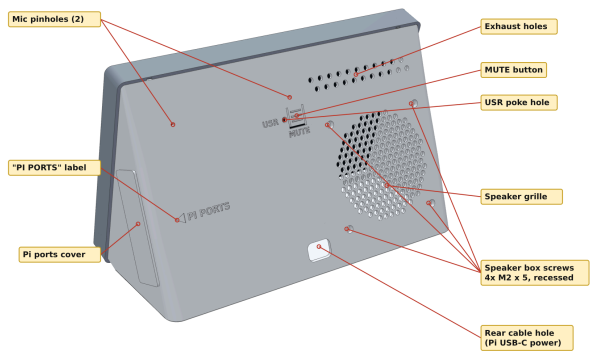


speaker-box: open rim down. The inside lid leans at 37°; it prints without supports.

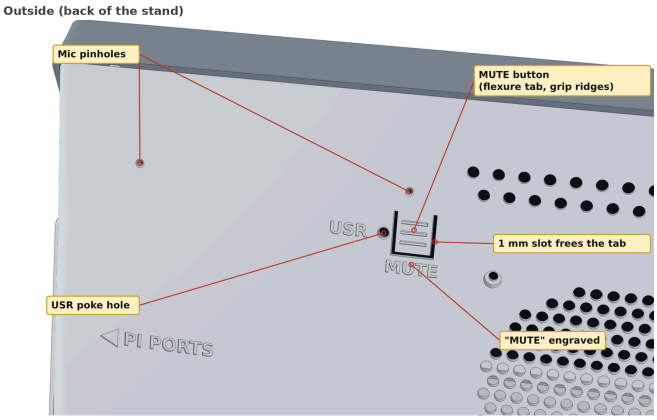


pi-ports-cover: flat plate down (the outside face), clip rim up.

Features to protect: please check these on the stand



Back of the stand (shown upright)



MUTE button area, close-up

Feature	Size	What to watch for
MUTE button (U-shaped slot)	1.0 mm slot, 9 mm tab	The tab must be free on three sides and flex when pressed. If strings or a skin bridge the slot, cut them clear with a hobby knife. Do not glue or fill it.
Mic pinholes (2)	Ø1.5 mm	Must be open right through. Clear them by hand with a 1.5 mm drill bit or pin. Don't enlarge them.
USR poke hole	Ø2.6 mm	Must be open right through.
Screw pilot holes (stand: 4 Pi + 2 ReSpeaker + 4 speaker; box: 4)	Ø1.6 mm	For M2 self-tapping screws. **Don't drill them out.**
Speaker screw recesses (4, back of stand)	Ø4.4 mm	Open from outside, with a flat seat inside for the screw head.
Speaker grille, exhaust holes	Ø3 mm	Clear any strings.
Engraved text (MUTE, USR, PI PORTS)	0.6 mm deep	Mirrored in the file on purpose, so it reads correctly on the finished part.
Snap tabs (front rim, 10) and frame grooves	—	Don't sand or trim them; they hold the frame on.
Upper Pi standoffs (2, inside)	Ø6.5 mm posts	The top edge on one side starts with a tiny overhang. If there's a whisker or blob, trim it so the top is flat.

Speaker box

- It must be **airtight**: 4 perimeters, no gaps or under-extrusion in the walls. Hold it up to a light to check for pinholes.
- The small hole on one side is the speaker wire exit, not a defect.

Before handing over (quick checklist)

- ☐ Brim removed, front rim edge clean
- ☐ MUTE tab flexes and springs back
- ☐ Both mic pinholes open (light shows through)
- ☐ USR hole, grille and vent holes open
- ☐ Frame snaps onto the stand's front rim[1]
- ☐ Cover clips into the stand's side opening[1]
- ☐ Speaker box: no gaps or pinholes in the walls
- ☐ No strings inside the stand

[1] A test fit is welcome but optional. Press gently: the tabs and clips are sized for a firm snap. Don't force them or trim anything if they're tight; just let the customer know.

Licence: personal-use print. This design is a remix of "Guition 4 inch Screen Frame and Stand" by MegaAndy (Printables), licensed CC BY-NC 4.0: not for resale. Source and assembly guide: github.com/drewballz1/Home-Assistant-Display-and-Satellite---Raspberry-Pi